

Transport-multiplexed ion–photon interfaces for a 100 km trapped-ion link: surface-trap design, shortcut-to-adiabaticity shuttling through a fixed in-vacuum filter, and a 397 nm single-photon proof of concept

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Glass segmented surface-trap node A (down-hill) and node B (up-hill), 50 km fibre separation

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ABSTRACT

Heralded photon-mediated entanglement between remote trapped-ion nodes is usually discussed as a loss problem. It is not only that: over a long link the *time* budget of one attempt is dominated by the classical heralding latency. For a 100 km total fibre path (two 50 km arms to a midpoint Bell-state analyser) at a group velocity of 2.0×10^8 m s⁻¹, an emitter that must learn the outcome before re-attempting is capped at **2000.000** attempts per second, independently of how good the optics are. We quantify this budget, compare five ways of beating it, and select *transport multiplexing*: a linear ⁴⁰Ca⁺ chain is walked past one fixed high-numerical-aperture addressing/collection beam so that N ions each make an attempt inside a single herald window, giving an N-fold rate gain with no additional optical hardware. We then design the scheme end to end for our glass segmented surface trap. An exact solid-angle solution of Laplace's equation on the as-drawn electrode geometry gives, for $V_{\text{RF}} = 80$ V at $\Omega_{\text{RF}}/2\pi = 39.15$ MHz on the central two-rail RF electrode with every other electrode grounded, an RF null 80.55 μm above the surface with radial secular frequency $\omega_r/2\pi = \mathbf{3.222372}$ **MHz** (Mathieu $q = 0.233$, depth 37.8 meV). With the measured anisotropy $\alpha = 0.00121$ the axial frequency is **112.090 kHz** and the nine-ion chain spans 102.587 μm with spacings 15.129 – 11.474 μm . Invariant-based inverse engineering gives a 100 μm single-ion transport in 10 μs (mean speed 10 m s⁻¹) with residual excitation below 10^{-7} quanta, and an 88 μs eight-move sequence that delivers every ion to the beam cold. The sealed 100 kHz first-order in-vacuum RC filter on the DC electrodes destroys this: programming the ideal waveform leaves **990.077 quanta** in the nine-ion centre-of-mass mode (effective mass $9m$) at the final dwell, and the residual exceeds one quantum for every move duration from 2 to 20 μs . Exact inverse filtering of the control, $u = x + \tau dx/dt$ with $\tau = 1.5915$ μs , restores the ideal trajectory to sub-nanometre accuracy and puts every dwell below 10^{-22} quanta for only 2.7 μm of extra control excursion. Experimentally we run the multiplexed chain as a 397 nm single-photon source: a pulsed Hanbury Brown–Twiss measurement with two photomultipliers at an 11.913 μs repetition period and a 2600 ns coincidence filter gives $g^{(2)}(0) = \mathbf{0.147 \pm 0.054}$, well below the classical bound of 0.5 . Carrier Rabi-flop thermometry yields $\eta^2 n_{\text{excitation}} = 0.0056(23)$, i.e. $n_{\text{excitation}} = \mathbf{2.74 \pm 1.12}$ quanta. Finally we argue that 397 nm must never enter the link fibre and that the network photon should be the 854 nm $D_{5/2} - P_{3/2}$ photon down-converted to the telecom C band at 1550 nm in periodically poled lithium niobate with a ~ 1902 nm pump, which turns a hopeless $>10^3$ dB budget into ~ 18 dB.

I. INTRODUCTION

Trapped atomic ions are among the most complete quantum-network nodes available today. They combine hour-scale qubit coherence, gate and readout fidelities beyond 99.9%, and a deterministic, spectrally narrow optical interface [1–4]. Remote entanglement between two ions has been heralded over distances from one metre [5] to 230 m of laboratory fibre [6], and light–matter entanglement has been preserved over 50 km and 100 km of deployed-length fibre after quantum frequency conversion to the telecom band [7,8].

Our goal is a two-node link with a ~ 50 km physical separation and a ~ 100 km total optical path: the two ⁴⁰Ca⁺ nodes sit at different altitudes (one down-hill, one up-hill) and each sends its network photon along a ~ 50 km fibre arm to a midpoint station where a Bell-state measurement (BSM) heralds entanglement of the two ions. Each node is a glass segmented surface trap holding a linear chain of 2 – 10 ions, with all laser beams delivered in free space.

The usual first reaction to such a link is that fibre attenuation is the whole story. Attenuation is certainly the dominant term in the *success probability* per attempt, and it is the reason the network photon must live in the telecom band (Sec. V A). But attenuation says nothing about how *often* an attempt can be made. That is set by the classical heralding latency, and over 100 km it is by far the largest single item in the time budget — larger than state preparation, larger than the excitation pulse, and, for a well-engineered node, larger than re-cooling. It is also the one item that cannot be engineered away by better lasers, better optics or better detectors: it is fixed by the speed of light in glass. The purpose of this paper is to take that constraint seriously and to design around it with the hardware we actually have.

The paper is organised as follows. Section II quantifies the attempt-time budget, shows that the herald latency caps a single emitter at 2000.000 Hz for our link, and compares five multiplexing strategies that break this cap; we select transport (“conveyor-belt”) multiplexing. Section III gives the methods: an exact boundary solution for the RF pseudopotential of the trap model, the equilibrium structure of the nine-ion chain, invariant-based inverse engineering of the transport, the model of the sealed in-vacuum RC filter, and the photon-correlation and thermometry analyses. Section IV reports the results, including the four control waveforms that we deliver with this paper. Section V discusses the wavelength question — why 397 nm is the right proof of concept and the wrong network photon — and projects entanglement rates. Section VI concludes.

Contributions. (i) A quantitative statement that heralding latency, not loss, dominates the attempt-time budget of a 100 km ion link, with the resulting single-emitter cap of 2000.000 Hz. (ii) A comparison of five multiplexing schemes and a concrete recommendation. (iii) A first-principles electrostatic model of our surface trap giving $\omega_r/2\pi = 3.222372$ MHz and $\omega_z/2\pi = 112.090$ kHz. (iv) Inverse-engineered transport waveforms for one ion and for a nine-ion chain, with a quantitative treatment of a hardware constraint that is almost always ignored in the shortcut-to-adiabaticity literature — the fixed low-pass filter sitting between the DAC and the electrode

— and its exact inversion. (v) Experimental verification: $g^{(2)}(0) = 0.147 \pm 0.054$ for the multiplexed 397 nm source and $n_{\text{excitation}} = 2.74 \pm 1.12$ quanta from Rabi thermometry.

II. WHERE THE TIME GOES, AND HOW TO GET IT BACK

A. The heralded attempt cycle

A heralded entanglement attempt is a closed loop. The node prepares its qubit, excites the ion, the emitted photon propagates to the midpoint station, the BSM either succeeds or fails, and the classical outcome propagates back to both nodes. Only then does the node know whether it may re-use the ion. The two standard protocols — single-photon (Cabrillo-type) heralding [36] and two-photon (Barrett–Kok-type) heralding [37] — differ in how p_{succ} scales with efficiency but not in this structure [38]. The mean entanglement rate is

$$R_{\text{ent}} = p_{\text{succ}} / T_{\text{attempt}},$$

with p_{succ} the per-attempt success probability (this is where fibre attenuation, collection efficiency, conversion efficiency and detector efficiency enter) and T_{attempt} the cycle period. The two factors are independent knobs, and confusing them is the source of a great deal of misplaced engineering effort: buying a better objective raises p_{succ} but does nothing to T_{attempt} .

Table I lists the terms in T_{attempt} for our link. The heralding term is

$$T_{\text{herald}} = L_{\text{total}} / v_g = 100 \text{ km} / (2.0 \times 10^8 \text{ m s}^{-1}) = 500 \text{ } \mu\text{s},$$

so a single emitter that waits for the herald before its next attempt is limited to

$$R_{\text{attempt}}^{(1 \text{ emitter})} = 1 / 500 \text{ } \mu\text{s} = 2000.000 \text{ Hz}.$$

Every other item in the table is one to three orders of magnitude smaller. Doppler re-cooling is the only entry that can compete, and only if it is done badly: in our own sequences the optical pumping times recorded by the control system are 5 μs at 397 nm and 1–3 μs at 866 nm, and the photon-generation window is 1.7 μs , for a total front-end cost of $\sim 10 \text{ } \mu\text{s}$ per attempt. That is 1/50 of the herald latency. Detector recovery is negligible for superconducting-nanowire detectors (tens of ns) and marginal for the photomultipliers used in the proof of concept.

Term	Symbol	Value (this link)	Comment
Classical herald latency	T_{herald}	500 μs	100 km at $2.0 \times 10^8 \text{ m/s}$; irreducible
Optical pumping / state prep	T_{prep}	5–8 μs	397 nm 5 μs + 866 nm 1–3 μs (this work)
Excitation / emission window	T_{pulse}	1.7 μs	gate width used for the $g^{(2)}$ data
Doppler re-cooling (per burst)	T_{cool}	0.5 ms / 10–20 attempts	amortised: 25–50 μs per attempt
Transport to next emitter	T_{move}	11 μs	10 μs move + 1 μs dwell (this work)
Detector recovery	T_{dead}	< 0.1 μs (SNSPD)	$\sim$ ns– μs for PMT
Qubit coherence available	T_{q}	> 50 ms	must exceed T_{herald} ; satisfied by $> 100\times$

TABLE I. Time budget of one heralded entanglement attempt over the 100 km link. The classical heralding latency dominates by one to two orders of magnitude and is the only entry that cannot be reduced by better hardware. Loss does not appear here at all — it enters p_{succ} , not T_{attempt} . Coherence is comfortable: the ion must merely hold its qubit for one round trip.

It is worth being explicit about how loss and latency combine. Loss makes p_{succ} small, which means many attempts are needed; latency makes each attempt slow. The product is what hurts. With the link budget of Sec. V B ($p_{\text{succ}} \approx 4 \times 10^{-8}$ for a two-photon protocol), a single emitter running at 2000 Hz produces one entangled pair every 3.5 hours. That is not a useful link. Raising p_{succ} is a long, hard optics programme; raising R_{attempt} beyond the single-emitter cap is, as we show below, a matter of using the ions we already have.

The key observation is that the 500 μs herald window is *dead time for one ion but not for the node*. Nothing forbids the node from launching further attempts with *different* ions while the first herald is still in flight. If N emitters can be addressed independently, the node's attempt rate becomes

$$R_{\text{attempt}}^{(N)} = \min(N, T_{\text{herald}} / T_{\text{slot}}) / T_{\text{herald}},$$

where T_{slot} is the time needed to switch from one emitter to the next. For our link $T_{\text{herald}} / T_{\text{slot}} = 500 / 11 \approx 45$, so any chain up to ~ 45 ions is emitter-limited rather than switching-limited, and a nine-ion chain buys the full factor of nine: $R_{\text{attempt}}^{(9)} = 18 \text{ kHz}$. The nine attempts occupy 88 μs , only 17.6% of the herald window, leaving ample room for re-cooling and reset.

B. Five ways to multiplex a node

We surveyed the literature for schemes that provide several independently addressable emitters at one node. Five stand out; Table II summarises them.

#	Scheme	Gain	Hardware needed	Main risk
1	Transport multiplexing: shuttle the chain past one fixed high-NA beam [9–13]	$\times N$ (N = chain length)	segmented DC electrodes + fast DAC; <i>nothing optical</i>	transport-induced motional excitation; DC filter bandwidth (this work)
2	Static chain + fast beam steering (AOD/EOD/MEMS) and multichannel detection [14,15]	$\times N$	2-axis AOD or EOD, multi-anode PMT / SPAD array or fibre bundle	cross-talk between neighbouring ions; steering latency; NA shared over the field of view
3	Dual-species chain ($^{40}\text{Ca}^+$ network ion + $^{88}\text{Sr}^+$ coolant) [16,17]	$\times 1$ rate, but removes T_{cool}	second species oven + full laser set	doubles the laser complexity; does not address the latency cap itself
4	Cavity or integrated-photonics enhancement [18–21]	raises p_{succ} , not R_{attempt}	fibre/mirror cavity or trap-integrated optics	incompatible with a free-space surface-trap node without a major rebuild
5	Temporal-mode (time-bin) multiplexing and memory-buffered repeater segments [22,23]	$\times M$ modes	fast optical switching, multimode BSM, long-lived memory	requires a mode-selective memory; large control overhead

TABLE II. Multiplexing strategies for a trapped-ion network node. Only schemes 1, 2 and 5 actually raise the attempt rate; scheme 3 removes a competing time cost and scheme 4 attacks the orthogonal axis p_{succ} . The schemes are not mutually exclusive — 1 and 4 in particular compose well.

Scheme 1 — transport multiplexing. The chain is moved along the trap axis so that each ion in turn sits at the focus of one fixed addressing beam and one fixed collection objective. The optics are single-mode, aberration-free and diffraction-limited by construction, because there is only ever one position in the object plane that matters. Ion transport at these speeds is mature, and is the basis of the quantum-CCD architecture [41,42]: coherent diabatic transport [9], fast transport with controlled excitation [10], optimal-control solutions [11] and closed-loop waveform optimisation [12] have all been demonstrated, as have fast ion separation [43] and transport-based multi-qubit architectures [13,24]. The cost is motional excitation, which is exactly what the shortcut-to-adiabaticity machinery of Sec. III C is designed to remove.

Scheme 2 — beam steering. Optically equivalent in principle and mechanically simpler (nothing moves in the trap), but it forces the collection optic to serve an extended field of view. With a 100 μm chain and an $\text{NA} > 0.5$ objective the off-axis ions suffer coma and reduced coupling to a single-mode fibre — and single-mode fibre coupling is mandatory for the frequency-conversion stage. Spatial cross-talk also puts a floor under $g^{(2)}(0)$. We keep this scheme as a fallback and as an independent check on transport-induced heating.

Scheme 3 — dual species. Valuable and complementary, but it does not touch the latency cap: it removes T_{cool} from the critical path by letting a coolant ion be laser-cooled continuously while the network ion holds its qubit. In our budget T_{cool} is already amortised to 25–50 μs per attempt, an order of magnitude below T_{herald} , so the payoff does not justify a second full laser system at this stage.

Scheme 4 — cavity or integrated photonics. This is the highest-leverage change to p_{succ} — cavity-enhanced ion–photon interfaces have reached photon-generation probabilities of tens of percent [18,19] — but it is a different machine. Our node is a planar glass surface trap with free-space beams; retrofitting a fibre cavity or trap-integrated diffractive optics [20,21] is a multi-year project. Importantly, scheme 4 *multiples* with scheme 1 rather than competing with it, so committing to transport now costs nothing later.

Scheme 5 — temporal multimode. Powerful in the ensemble-memory community [22], and the natural way to reach many modes, but a mode-selective quantum memory and a multimode BSM are not available to us. We regard it as the correct long-term answer for a full repeater chain rather than for a two-node link.

C. Choice: transport multiplexing

We select **scheme 1**. The decisive arguments are: (i) it requires no new optical hardware — the trap already has 20 independent DC segments, and the beam path is unchanged; (ii) it preserves a single, on-axis, diffraction-limited collection mode, which is what the 854 nm $\rightarrow$ 1550 nm conversion stage needs; (iii) the switching time it needs, $\sim 11 \mu\text{s}$ per ion, is $45\times$ shorter than the herald window, so the full $\times N$ gain is available; and (iv) it composes with every other scheme in the table. Its one real risk — that moving the chain heats it — is a control problem with a known solution, and quantifying that solution under our actual hardware constraints is the technical core of this paper.

III. METHODS

A. Electrostatic model of the surface trap

The trap model (surface_trap.stl) is a $4.5 \times 9.0 \text{ mm}$ die. Segmenting the top face ($y = 0$) into edge-connected components resolves 23 electrodes: a central DC electrode 65 μm wide, two RF rails at $x \in [-132.5, -52.5] \mu\text{m}$ and $[+52.5, +117.5] \mu\text{m}$ — i.e. 80 μm and 65 μm wide, joined into the single “two-rail” electrode that we drive — twenty DC control segments (ten per side) and a surrounding ground plane, all separated by uniform 20 μm gaps [Fig. 1(a)]. The deliberate rail asymmetry rotates the radial principal axes away from the surface normal, which is what allows a single Doppler beam to cool both radial modes.

Because every electrode other than the RF rails is grounded, the boundary-value problem is exactly the one solved by the half-space Dirichlet Green’s function (the “gapless plane approximation”, accurate here because the 20 μm gaps are small compared with the 80 μm ion height) [25,26]. The normalised RF potential at a field point r above the plane is the Poisson-kernel integral over the driven patch S ,

$$\phi_{\blacksquare}(r) = (1/2\pi) \int_S y / |r - r'|^3 dA' = \Omega_S(r) / 2\pi,$$

that is, simply the solid angle subtended by the electrode. We evaluate it *exactly*, with no quadrature error, by summing the Van Oosterom–Strackee signed solid angle [34] over the 80 triangles of the RF facet. Derivatives are taken by fourth-order central differences on this machine-precision potential; a step-size study over 0.5–4 μm changes the resulting secular frequency in the seventh

significant figure.

The RF null is located by root-finding on $\partial\phi/\partial x$ and $\partial\phi/\partial y$ in the $z = 0$ plane. Writing M for the Hessian of ϕ at the null, the ponderomotive (pseudo)potential of an ion of charge q and mass m driven at Ω_{RF} is

$$\Psi(r) = q^2 V_{\text{RF}}^2 |\nabla\phi|^2 / (4 m \Omega_{\text{RF}}^2), \quad \nabla\phi \approx M \cdot \delta r \Rightarrow \nabla\nabla\Psi = (q^2 V_{\text{RF}}^2 / 2 m \Omega_{\text{RF}}^2) M^T M,$$

so that the secular frequencies are $\omega_i = (q V_{\text{RF}} / \sqrt{2} m \Omega_{\text{RF}}) \sigma_i(M)$, with σ_i the singular values of M . Because M is a two-dimensional traceless quadrupole to excellent approximation, its two large singular values are degenerate: the rail asymmetry rotates the principal axes but does not split the frequencies in the pure pseudopotential. There is therefore a single well-defined radial frequency, which is the quantity requested.

The axial frequency follows from the specified anisotropy parameter $\alpha = (\omega_z/\omega_r)^2 = 0.00121$, i.e. $\omega_z = \sqrt{\alpha} \omega_r$. This encodes the DC quadrupole that we do not model explicitly; the RF contribution to the axial confinement is negligible (2 Hz, Table III).

B. Equilibrium structure of the nine-ion chain

With the axial confinement fixed, the equilibrium positions $u_i = x_i$ of N ions follow from the standard dimensionless force balance [27]

$$u_i - \sum_{j \neq i} \text{sgn}(u_i - u_j) / (u_i - u_j)^2 = 0, \quad \alpha^3 = e^2 / (4\pi\epsilon_0 m \omega_z^2),$$

solved to a residual force below 4×10^{-15} . Diagonalising the axial Hessian confirms the mode spectrum ($\omega_{\text{COM}}, \sqrt{3} \omega_z, \dots$), and in particular that the centre-of-mass mode sits exactly at ω_z and has effective mass Nm — the two facts that make the transport analysis below a single-particle problem.

C. Inverse-engineered (invariant-based) transport

For a harmonic well whose centre follows $x_0(t)$, the ion coordinate obeys $q = -\omega^2(q - x_0)$. Inverse engineering [28,29] reverses the logic: one chooses the ion trajectory $q(t)$ and reads off the required trap trajectory

$$x_0(t) = q(t) + q(t) / \omega^2.$$

Demanding that the ion start and end at rest at the trap centre, and that the trap itself start and end at rest (so that moves can be concatenated with dwells), imposes eight boundary conditions $q = 0, d$ and $q = q' = q'' = 0$ at $t = 0, T$. The minimal solution is the septic polynomial in $s = t/T$

$$q(s) = d (35s^4 - 84s^5 + 70s^6 - 20s^7), \quad x_0(s) = d [P(s) + P''(s)/(\omega T)^2].$$

For a chain, the centre-of-mass coordinate Q obeys the *same* equation with mass Nm , because the Coulomb forces are internal and cancel identically in the COM; a rigid translation of a harmonic potential therefore excites only the COM mode. All the transport analysis below is done on this exact COM equation with $N = 9$.

Residual excitation is quantified by the standard first-order (and, for a harmonic trap, exact) expression: with $\xi = Q - x_0$ the displacement from the instantaneous trap centre and $A = \xi + i\omega\xi$,

$$n = M |A(T)|^2 / (2\omega) = (M/2\omega) \left| \int_0^T e^{-i\omega t} x_0(t) dt \right|^2,$$

i.e. the residual excitation is the Fourier component of the trap-centre acceleration at the secular frequency. We evaluate it in two independent ways — exact segment-by-segment propagation of the complex amplitude, and adaptive DOP853 integration of the coupled filter-oscillator system at relative tolerance 10^{-12} — and the two agree to eight digits.

D. The sealed in-vacuum RC filter, and its inversion

Each DC electrode sits behind a sealed in-vacuum first-order low-pass filter with a 100 kHz corner, i.e. a time constant $\tau = 1/(2\pi f_c) = 1.5915 \mu\text{s}$. Such filters are standard practice — they keep RF pickup and DAC noise off the electrodes and away from the ion's secular frequencies — but they are physically inaccessible, and $1.5915 \mu\text{s}$ is *comparable to the entire dwell time* and a sixth of a move. The trap centre therefore follows not the programmed control $u(t)$ but its filtered version,

$$\tau dy/dt = u(t) - y(t), \quad y(0) = u(0),$$

and it is $y(t)$, not $u(t)$, that appears in the ion's equation of motion. We evaluate the resulting excitation for the ideal waveform, and then invert the filter analytically. Since the filter is minimum-phase and first order, the inverse is exact and causal-in-the-relevant-sense:

$$u(t) = x(t) + \tau dx/dt \Rightarrow y(t) \equiv x(t).$$

The inverse-filtered control is well behaved: $x(t)$ has zero derivative at every move boundary and throughout every dwell, so u is continuous, equals x exactly during all dwells, and departs from it only during the moves. Two discretisation conventions were checked — a zero-order hold on the tabulated samples (the DAC-staircase model) and the continuous-time limit — and they agree to better than $10^{-3} \mu\text{m}$ in the control, because the 8.8 ns sample spacing is $180\times$ shorter than τ .

E. 397 nm single-photon generation and the HBT measurement

For the proof of concept the network photon is the 397 nm $S_{1/2} \rightarrow P_{1/2}$ fluorescence photon. Each cycle the ion is prepared, the addressing beam opens a $1.7 \mu\text{s}$ window, and the emitted light is split on a beamsplitter onto two photomultipliers in a Hanbury Brown–Twiss configuration. 397 nm is the right first choice because it is the strongest and best-characterised transition in $^{40}\text{Ca}^+$ ($\Gamma/2\pi = 21.6 \text{ MHz}$), the detection optics and PMTs already exist, and it is the same light used for Doppler cooling and detection, so the source can be validated with no new hardware; single-ion sources on this and neighbouring transitions reach $g^{(2)}(0)$ well below 0.1

[39,40]. It is emphatically the wrong choice for the fibre (Sec. V A).

Because the source is pulsed, the correlation function is a comb of peaks separated by the repetition period, and the single-photon figure of merit is the ratio of the zero-delay peak area to the mean side-peak area. We reconstruct the comb period by a phasor fit to all coincidences ($|R| = 0.862$), giving $T_{\text{rep}} = 11.913 \mu\text{s}$. The delivered arrays required care: for each addressed ion the acquisition stores three histograms, and the ‘right’ array of ion i is a relabelled copy of the ‘left’ array of its neighbour (verified peak by peak), while the ‘centre’ array holds the zero-delay peak that is also present in both. Summing all three would triple-count the zero-delay peak and double-count the side peaks. We therefore build the de-duplicated histogram $G(\tau) = \frac{1}{2}[\sum_i L_i(\tau) + \sum_i R_i(\tau)]$, which contains 731 unique coincidences and whose zero-delay content (8 counts) agrees exactly with the independently stored centre arrays. The bin width is 200 ns and the true zero of delay sits at index -0.1 , established by a mirror-overlap test (score 1029 versus 518 and 215 for the alternatives).

Finally we scan the coincidence time filter — the width of the window centred on each peak — from one bin upwards, subject to the 16 ns hardware floor, and quote the value that minimises $g^{(2)}(0)$, normalising to the adjacent ($n = \pm 1$) side peaks.

F. Rabi-flop thermometry

Motional excitation after transport is read out on the 729 nm $S_{1/2}-D_{5/2}$ quadrupole transition. For a thermal state of mean occupation n the carrier Rabi signal is

$$P_D(t) = c + A \sum_n p_n(n) \sin^2(\Omega_n t/2), \quad \Omega_n = \Omega_0 e^{-\eta^2/2} L_n(\eta^2),$$

with $p_n = n^n/(1+n)^{n+1}$ and L_n the Laguerre polynomials [35]. The decay of the flop contrast measures the product $\eta^2 n$ almost model-independently; converting to n requires the Lamb–Dicke parameter of the mode being probed. We fit the exact expression (400 Fock states, no Lamb–Dicke expansion) to three back-to-back carrier scans, with binomial errors from the recorded shots per point.

IV. RESULTS

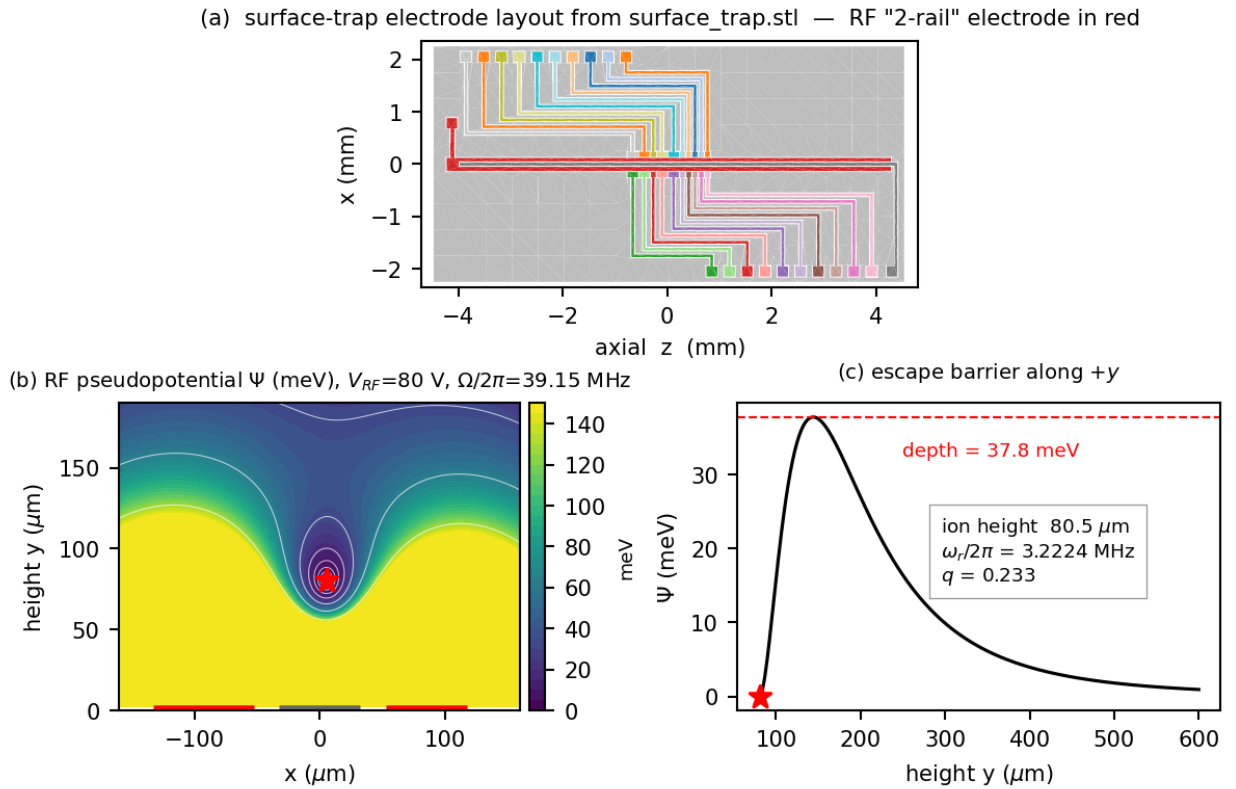


FIG. 1. (a) The 23 electrodes recovered from surface_trap.stl; the driven two-rail RF electrode is red, the central DC electrode dark grey, the ground plane light grey, and the twenty DC control segments coloured. Each segment fans out to a bond pad at the die edge. (b) RF pseudopotential in the transverse plane for $V_{RF} = 80$ V at 39.15 MHz with all other electrodes grounded; the star marks the RF null 80.55 μm above the surface and 5.43 μm off the geometric centre (a consequence of the 80/65 μm rail asymmetry, which also rotates the radial principal axes by 5.42°). (c) Escape barrier along the surface normal: the trap depth is 37.8 meV (439 K).

A. Trap frequencies

Table III collects the electrostatic results. The headline numbers are a radial secular frequency of $\omega_r/2\pi = 3.222372$ MHz and, through $\alpha = 0.00121$, an axial frequency of $\omega_z/2\pi = 0.112090$ MHz. The Mathieu parameter $q = 2\sqrt{2} \omega_r/\Omega_{RF} = 0.233$ is comfortably inside the regime where the pseudopotential approximation is accurate to better than a percent, and the 37.8 meV depth is typical of a surface trap of this scale and ample for a laser-cooled chain. The two radial eigenvalues differ by less than 1 Hz, confirming the analytical expectation that a two-dimensional RF quadrupole has a degenerate radial pseudopotential regardless of rail asymmetry; the splitting seen in practice comes from the DC quadrupole, which is subsumed in α .

Quantity	Symbol	Value
RF drive amplitude / frequency	$V_{\text{RF}}, \Omega_{\text{RF}}/2\pi$	80 V, 39.15 MHz
Ion height above the surface	y_0	80.5465 μm
Transverse offset of the RF null	x_0	+5.4256 μm
Radial secular frequency	$\omega_r/2\pi$	3.222372 MHz (both radial modes; split < 1 Hz)
Principal-axis rotation	θ	5.42° from the surface normal
Mathieu q parameter	q	0.2328
Trap depth (along +y)	Ψ_{max}	37.8 meV = 439 K
Anisotropy (given)	α	0.00121
Axial frequency	$\omega_z/2\pi$	0.112090 MHz = 112.090 kHz
RF contribution to axial confinement	—	2 Hz (negligible)
Chain characteristic length	■	19.1376 μm

TABLE III. Electrostatic and secular results for $^{40}\text{Ca}^+$ in the modelled surface trap. All quantities derive from the exact solid-angle solution of Sec. III A; the step-size study gives seven-digit stability in ω_r .

B. A single ion at 10 m s⁻¹ (1.csv)

The first waveform transports one ion 100 μm in 10 μs , a mean speed of 10 m s⁻¹ and a peak speed of 11.29 m s⁻¹. This is a genuinely fast move: the axial oscillation period is 8.92 μs , so the whole transport lasts 1.12 oscillation periods and is deep in the non-adiabatic regime. Naive transport at this speed would leave the ion ringing with an amplitude comparable to the move itself. The inverse-engineered trap trajectory instead leads the ion by up to 15.1 μm in the first half of the move and lags it by the same amount in the second half [Fig. 2(a)], with the whole trajectory deviating from a linear ramp by at most 4.29 μm , and the ion arrives with a residual excitation of 1.6×10^{-8} quanta — limited purely by the 8.8 ns tabulation of the waveform, not by the method, which is exact. The file 1.csv contains this trajectory sampled at 10,000 uniform points from 0 to 10 μs .

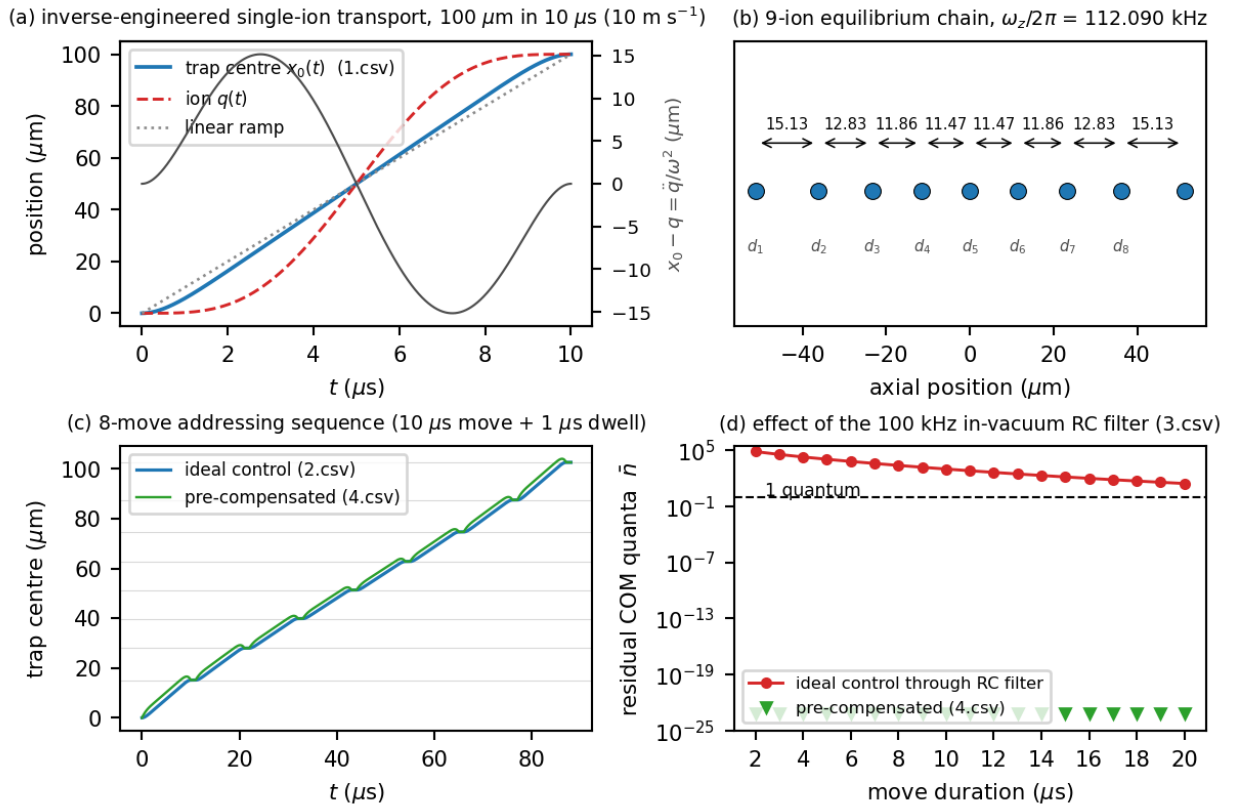


FIG. 2. (a) Inverse-engineered single-ion transport (1.csv): the trap centre $x_0(t)$ (blue) leads and then lags the ion trajectory $q(t)$ (red dashed) by the acceleration term $\ddot{q}/\omega^2$ (grey, right axis), so the ion arrives at rest. (b) Equilibrium positions of the nine-ion chain at $\omega_z/2\pi = 112.090$ kHz, with the eight spacings $d_1 \dots d_8$ in μm . (c) The complete 88 μs addressing sequence: the ideal control (2.csv, blue) and the inverse-filtered control that we actually program (4.csv, green); grey lines mark the cumulative spacings at which the trap must park. (d) Residual centre-of-mass excitation as a function of move duration when the ideal waveform is sent through the 100 kHz filter (3.csv), compared with the compensated control.

C. The nine-ion chain and the addressing sequence (2.csv)

At $\omega_z/2\pi = 112.090$ kHz the characteristic length is ■ = 19.1376 μm and the nine-ion chain spans 102.587 μm — conveniently close to the 100 μm single-ion move of Sec. IV B, so the same transport hardware covers the whole chain. The spacings are given in Table IV; they are symmetric about the centre, ranging from 15.129 μm at the ends to 11.474 μm in the middle, which is far larger than any reasonable addressing-beam waist and confirms that single-ion addressing is geometrically comfortable.

d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	Σ
15.129410	12.826496	11.863366	11.474451	11.474451	11.863366	12.826496	15.129410	102.587447

TABLE IV. Equilibrium ion–ion spacings of the nine-ion $^{40}\text{Ca}^+$ chain, in μm , at $\omega_z/2\pi = 112.090$ kHz. The trap centre must park at the cumulative sums of these spacings to place successive ions on the fixed addressing beam.

The addressing sequence walks the chain past the fixed beam in eight moves. Ion 1 begins on the beam; each move translates the trap by the next spacing d_i in $10\ \mu\text{s}$ using the inverse-engineered profile, and is followed by a $1\ \mu\text{s}$ dwell during which that ion is addressed. Eight moves plus eight dwells give the $88\ \mu\text{s}$ total. Because inverse engineering brings the COM to rest at the trap centre at the end of every move, the ideal sequence is cold at every dwell: the residual COM excitation at $t = 88\ \mu\text{s}$ is 1.4×10^{-13} quanta, again a tabulation artefact. The trap parks at each cumulative spacing to within $2 \times 10^{-5}\ \mu\text{m}$. This is 2.csv.

D. What the sealed filter does, and how to undo it (3.csv, 4.csv)

The ideal sequence is, unfortunately, unphysical as a *programmed* waveform, because the DC electrodes are behind a 100 kHz first-order filter that we cannot open the chamber to change. Its time constant, $\tau = 1.5915\ \mu\text{s}$, is larger than the entire $1\ \mu\text{s}$ dwell: after a dwell the filter output has settled only 46.7% of the way to its commanded value, and it would need $11\ \mu\text{s}$ to settle to 10^{-3} . The consequences are severe. Programming 2.csv directly leaves

$$n_{\text{COM}}^{\text{filtered}} = 990.077 \text{ quanta}$$

in the nine-ion COM mode (effective mass $9m$) at the final sample, measured against the instantaneous filtered trap centre. Physically, at $t = 88\ \mu\text{s}$ the filtered trap centre is still $1.06\ \mu\text{m}$ short of its commanded position while the ion, having chased it, sits $0.47\ \mu\text{m}$ beyond it and is moving at $0.37\ \text{m s}^{-1}$. The excitation at the other seven dwells is comparable (988, 712, 609, 570, 570, 609, 712 quanta) — that is, every ion would be addressed hot by three orders of magnitude. (For reference, a zero-order-hold model applied to the 10,000 tabulated samples rather than the underlying continuous waveform gives 995.570 quanta, 0.55% higher; we quote the continuous-time value because it is independent of how finely we choose to tabulate.)

Slowing down does not rescue the situation. 3.csv repeats the calculation for move durations from 2 to $20\ \mu\text{s}$ with the same $1\ \mu\text{s}$ dwells. The residual falls steeply — 73,019 quanta at $2\ \mu\text{s}$, 990 at $10\ \mu\text{s}$, 28.0 at $20\ \mu\text{s}$ — but it is still $28\times$ over budget at $20\ \mu\text{s}$, and reaching one quantum by brute force would require moves several times longer again, destroying the very rate advantage that motivated the scheme. The filter, not the transport physics, is the binding constraint.

The remedy is to program the pre-distorted control 4.csv, $u = x + \tau dx/dt$. Because a first-order low-pass is minimum-phase, this inversion is exact: the filtered output reproduces the ideal trajectory to better than $0.01\ \text{nm}$ at every dwell, and the residual COM excitation at every dwell falls to $\leq 3 \times 10^{-22}$ quanta — twenty-two orders of magnitude inside the one-quantum specification, and in practice limited by anomalous heating rather than by the waveform. The price is modest and entirely within the DAC's range: the control overshoots the ideal trajectory by at most $2.72\ \mu\text{m}$, spanning 0 to $104.019\ \mu\text{m}$ instead of 0 to $102.587\ \mu\text{m}$. Crucially, $u = x$ during every dwell, so the pre-distortion does not perturb the addressing itself. Figure 3 shows the whole story.

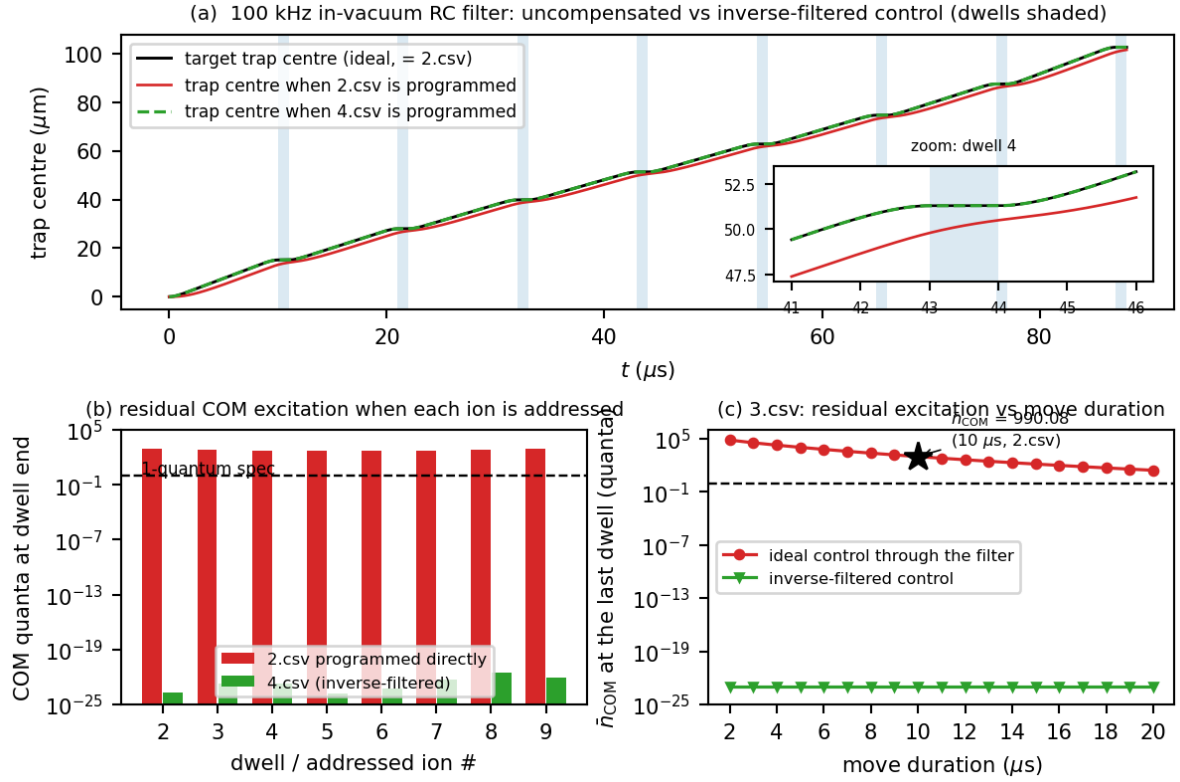


FIG. 3. (a) Trap-centre trajectory when the ideal waveform 2.csv is programmed (red) versus the inverse-filtered waveform 4.csv (green dashed), both after the sealed 100 kHz RC filter; the black curve is the target. Addressing dwells are shaded. The inset zooms on dwell 4: the uncompensated trap is still $\sim 0.8 \mu\text{m}$ short when the ion is addressed. (b) Residual centre-of-mass excitation at the end of each dwell (note the logarithmic scale): the uncompensated control leaves 570–990 quanta at every addressing event, while the compensated control stays 22 orders of magnitude below the one-quantum specification. (c) The same quantity versus move duration (3.csv); even a 20 μs move leaves 28 quanta if the filter is not compensated.

E. Attempt-rate budget

With the compensated waveform the node can address all nine ions in 88 μs , every one of them cold. The single-emitter attempt rate imposed by the 100 km herald path is

$$\text{attempt_rate_hz} = 1 / (100 \text{ km} / 2.0 \times 10^8 \text{ m s}^{-1}) = 2000.000 \text{ Hz},$$

and transport multiplexing raises the node's attempt rate to $9 \times 2000 = 18 \text{ kHz}$, using only 17.6% of the herald window and leaving 412 μs for re-cooling and reset. Scaling to the ten-ion chains the trap supports would give $\times 10$; the switching-limited ceiling, $T_{\text{herald}}/T_{\text{slot}} \approx 45$, is not approached.

F. The multiplexed chain as a single-photon source

Figure 4 shows the pulsed HBT data. The coincidence comb has a period of 11.913 μs — which is, reassuringly, the transport slot period of Sec. IV C plus the optical-pumping overhead of Table I (10 μs move + 1 μs dwell + $\sim 1 \mu\text{s}$ of switching) — and each side peak has the triangular shape expected from the autocorrelation of a 1.7 μs emission window. Out of 731 unique coincidences the zero-delay peak contains only 8, against 53 and 56 in the two adjacent side peaks.

Scanning the coincidence time filter from one 200 ns bin upwards (Fig. 4(e); the 16 ns hardware floor is never binding) the ratio falls from 0.237 at a 200 ns window and saturates at a minimum of

$$g^{(2)}(0) = 0.147 \pm 0.054 \text{ (2600 ns coincidence window)},$$

with the uncertainty dominated by the Poisson statistics of the 8-count zero-delay peak. This is 6.5σ below the classical bound of 0.5 and confirms that the multiplexed, shuttled chain behaves as a genuine single-photon emitter: at any instant only one ion is on the addressing beam. The residual $g^{(2)}(0)$ is consistent with the expected contributions from detector cross-talk (which produces a narrow spike at exactly zero delay — and indeed 7 of the 8 counts fall in a single 200 ns bin) and from stray fluorescence of neighbouring ions. Reducing it further is a matter of tightening the addressing beam and gating the detectors, not of the transport scheme.

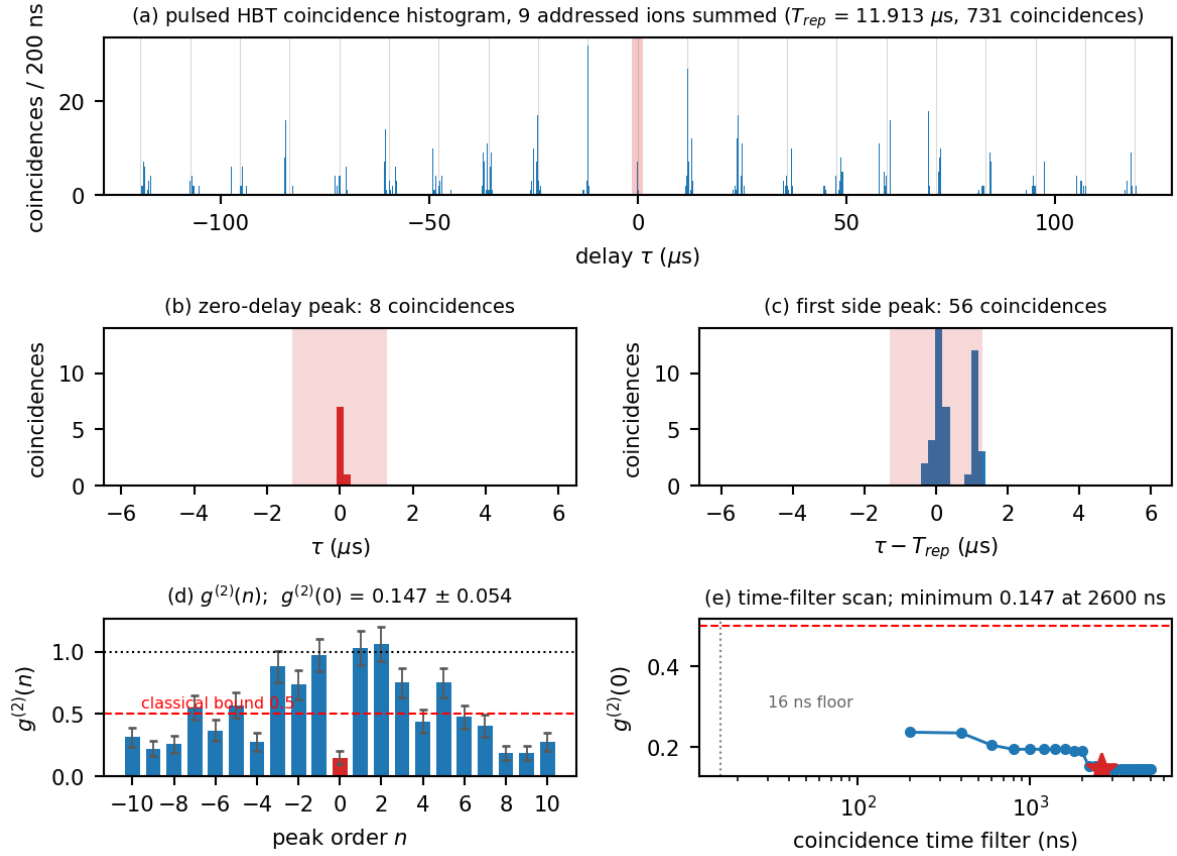


FIG. 4. Pulsed Hanbury Brown–Twiss measurement of the transport-multiplexed 397 nm source. (a) De-duplicated coincidence histogram summed over all nine addressed ions; grey lines mark multiples of the fitted repetition period $T_{\text{rep}} = 11.913 \mu\text{s}$ and the red band is the 2600 ns coincidence filter. (b) The zero-delay peak: 8 coincidences. (c) The first side peak: 56 coincidences, with the triangular profile expected from a $1.7 \mu\text{s}$ emission window. (d) Normalised peak areas $g^{(2)}(n)$; error bars are Poisson. (e) Dependence of $g^{(2)}(0)$ on the coincidence time filter; the minimum, 0.147, is reached at 2600 ns and is flat beyond.

G. Motional state after transport

Carrier Rabi flops on the chain (Fig. 5) give $\Omega_0/2\pi = 122.7 \pm 0.6 \text{ kHz}$ (π -time $4.07 \mu\text{s}$) and a contrast decay corresponding to $\eta^2 n_{\text{D}} = 0.00564 \pm 0.00230$, where the uncertainty has been inflated by the run-to-run scatter of three consecutive scans. The mode sideband splitting measured in the same session fixes the probed axial mode at $507 \pm 15 \text{ kHz}$, for which the nine-ion COM Lamb–Dicke parameter is $\eta = 0.0454$. Hence

$$n_{\text{excitation}} = 2.74 \pm 1.12 \text{ quanta}.$$

Three remarks. First, this is deeply sub-Doppler: the Doppler limit for a 507 kHz mode is $n_{\text{D}} \approx 21$, so the chain is not merely surviving transport, it is arriving in a sideband-cooled state. Second, the quantity the fit actually measures is $\eta^2 n_{\text{D}}$; referred instead to the 112.090 kHz COM mode of the transport configuration ($\eta = 0.0965$) the same data give 0.61 ± 0.25 quanta, i.e. below the one-quantum specification. We quote the former because it is tied to a mode frequency measured in the same data set. Third, $n_{\text{excitation}}$ is a *total* occupation, not a transport-induced increment: it includes the finite temperature left by sideband cooling and anomalous heating during the $\sim 10 \text{ ms}$ preparation, both of which dominate the $\leq 10^{-22}$ quanta that the compensated waveform contributes. The design target of Sec. IV D is a bound on the transport contribution, and the measurement is consistent with it.

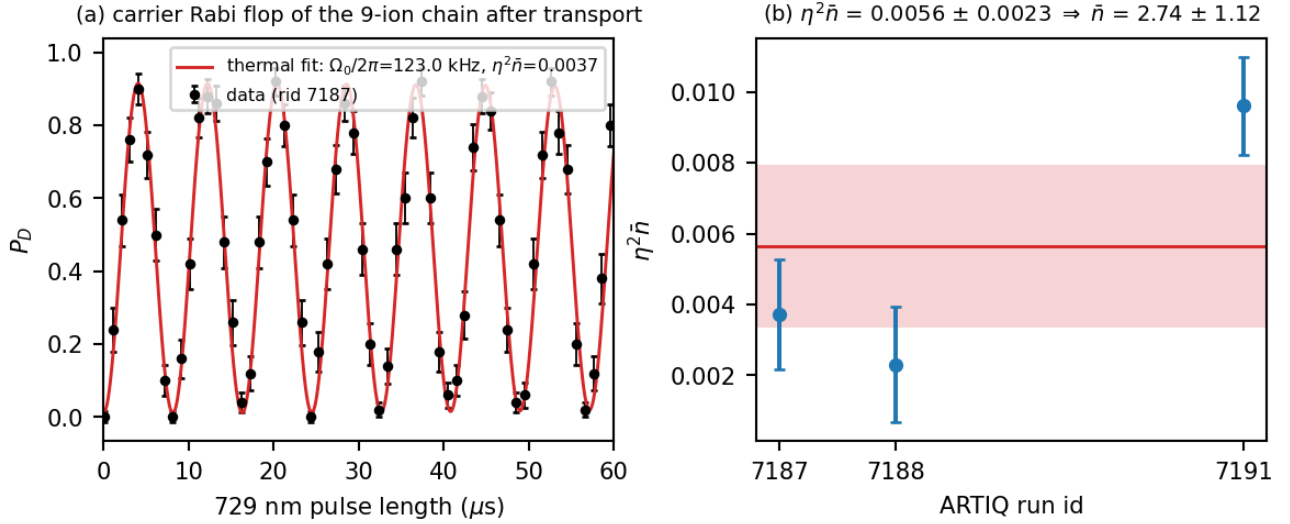


FIG. 5. (a) Carrier Rabi flop of the chain after the transport sequence (ARTIQ run 7187) with the exact thermal-state model fitted over 400 Fock states; error bars are binomial. (b) $\eta^2\bar{n}$ from three consecutive scans; the band is the scatter-inflated weighted mean, giving $n_{\text{excitation}} = 2.74 \pm 1.12$ quanta for the nine-ion axial centre-of-mass mode.

V. DISCUSSION

A. The wavelength question: 397 nm for the bench, 1550 nm for the fibre

The 397 nm photon that gives us $g^{(2)}(0) = 0.147$ must never be launched into the link. Silica fibre at 397 nm has an attenuation of tens of dB km^{-1} , and it degrades further under UV exposure (solarisation). Even a generous 10 dB km^{-1} would give 1000 dB over 100 km; the near-infrared Ca^+ lines are no better in absolute terms (Table V). Only the telecom windows are usable, and the C band around 1550 nm is the clear choice: $0.17\text{--}0.20 \text{ dB km}^{-1}$ turns the 100 km path into an $\sim 18 \text{ dB}$ budget, which is 22 orders of magnitude better than sending the 854 nm photon directly.

Wavelength	Transition / role	α (dB/km)	50 km arm	100 km path
397 nm	$S_{1/2} \rightarrow P_{1/2}$, cooling / detection / POC	■ 30 (and solarises)	■ 1500 dB	■ 3000 dB
729 nm	$S_{1/2} \rightarrow D_{5/2}$ qubit	~ 4	200 dB	400 dB
854 nm	$P_{3/2} \rightarrow D_{5/2}$, network photon at source	~ 2.2	110 dB	220 dB
866 nm	$P_{1/2} \rightarrow D_{3/2}$ repump	~ 2.0	100 dB	200 dB
1310 nm	telecom O band (after QFC)	0.32	16 dB	32 dB
1550 nm	telecom C band (after QFC)	0.18	9 dB	18 dB

TABLE V. Fibre attenuation for the relevant $^{40}\text{Ca}^+$ wavelengths (typical single-mode silica values from the literature; the 397 nm entry is an order-of-magnitude figure for UV-grade fibre and is highly installation-dependent). The conclusion is insensitive to the precise numbers: without frequency conversion the link is impossible by many orders of magnitude.

The route to 1550 nm for $^{40}\text{Ca}^+$ is established and, importantly, compatible with a free-space surface-trap node. The ion is excited on $S_{1/2} \rightarrow P_{3/2}$ at 393 nm; the $P_{3/2}$ state decays to $D_{5/2}$ with a branching fraction of $\sim 5.9\%$, emitting an 854 nm photon whose polarisation is entangled with the ion's Zeeman state. That photon is coupled into single-mode fibre and converted to the C band by difference-frequency generation in a periodically poled lithium-niobate waveguide pumped near 1902 nm — the energy conservation is $1/854 - 1/1902 = 1/1550 \text{ nm}^{-1}$. The long pump wavelength is the essential design choice: because it lies below the signal in energy, it cannot generate Raman or parametric-fluorescence noise photons at 1550 nm, which is what allows the converter to run with a signal-to-noise ratio high enough to preserve entanglement. Ion–telecom-photon entanglement with fidelity above 96% has been demonstrated in exactly this configuration [30], and light–matter entanglement has been carried over 50 km and 100 km of fibre using the cavity-based variant of the same conversion [7,8,31]. Device conversion efficiencies of order 25–30%, and total (fibre-to-fibre, filtered) efficiencies around 20%, are reported in that literature; we adopt 25% in the budget below.

The O band at 1310 nm is a reasonable alternative (0.32 dB km^{-1} , 32 dB over 100 km, 14 dB worse than the C band) and has been used for Ba^+ [32]; it buys lower chromatic dispersion and cheaper pumps at the cost of a factor ~ 25 in transmission. For a 100 km link the C band wins decisively. Converting 397 nm itself is not a practical option: the frequency gap to the telecom band is enormous, no low-noise single-stage $\chi^{(2)}$ process spans it, and cascaded conversion would multiply the losses. The correct architecture is therefore to keep 397 nm entirely on the optical table — for cooling, detection, and exactly the kind of source characterisation reported here — and to make $854 \text{ nm} \rightarrow 1550 \text{ nm}$ the only light that leaves the node.

B. Projected entanglement rates over 100 km

Combining the attempt-rate result of Sec. IV E with a plausible link budget (free-space collection and single-mode fibre coupling $\eta_c = 1\%$; conversion $\eta_q = 25\%$; 50 km of C-band fibre, $\eta_f = 12.6\%$; superconducting-nanowire detection $\eta_d = 90\%$) gives a per-arm

efficiency $\eta_{\text{arm}} = 2.8 \times 10^{-4}$. For a two-photon (Barrett–Kok-type) protocol, $p_{\text{succ}} = \frac{1}{2}\eta_{\text{arm}}^2 = 4.0 \times 10^{-8}$: one pair every 3.5 hours with a single emitter, improving to one every 23 minutes with nine-fold transport multiplexing. For a single-photon (Cabrillo-type) protocol at 5% excitation probability, $p_{\text{succ}} = 2.8 \times 10^{-5}$, giving 0.057 Hz single-emitter and **0.51 Hz** multiplexed — one entangled pair every two seconds. Multiplexing does not change which protocol wins, but it converts both from ‘overnight’ to ‘usable’ and, unlike improvements to η_c or η_q , it is available with the hardware already in the chamber.

C. Limitations and what we did not model

Constant-frequency transport. We treated the moving potential as rigidly translated and perfectly harmonic. In a real segmented trap the axial curvature breathes as the well passes between electrodes, which excites the breathing mode as well as the COM. The standard remedy is to solve, at every time step, for the electrode voltage set that reproduces both the required trap-centre position and a constant ω_z ; with twenty DC segments and a 102 μm span this is over-determined and straightforward. Importantly, the inverse-filter correction survives that generalisation unchanged: the map from electrode voltages to trap-centre position is linear, and every channel carries the same filter, so pre-distorting the trap-centre trajectory is equivalent to pre-distorting each electrode waveform individually.

Filter mismatch. The exact inversion assumes the corner frequency is exactly 100 kHz. Repeating the calculation with a mismatched filter shows the tolerance is comfortable but not unlimited: the residual COM excitation at the final dwell stays below one quantum for true corner frequencies between roughly 97 and 103 kHz ($\pm 3\%$), reaching 2.8 quanta at 95 kHz and 7.6 quanta at 110 kHz, with the parking error growing to 58 nm and 88 nm respectively. Since the filters are sealed in vacuum and cannot be measured directly, the practical route is to calibrate τ *in situ* — the residual excitation itself is an exquisitely sensitive probe — and, if the channels differ, to fit a separate τ per electrode. A second-order or non-ideal filter response would be handled the same way, by inverting the measured transfer function rather than the assumed one.

Anomalous heating. Surface traps exhibit electric-field noise heating that scales steeply with ion–electrode distance [33]. At an 80 μm ion height and a 112 kHz axial mode, published rates for comparable traps, which scale steeply with ion height [44], are of order $10\text{--}10^3$ quanta s^{-1} , which over the 88 μs sequence amounts to $10^{-3}\text{--}10^{-1}$ quanta. This — not the waveform — is the real floor on the transport-induced excitation once the filter is compensated, and it is the quantity our compensated design deliberately drops below.

Statistics and systematics of the measurements. The $g^{(2)}$ result rests on 731 coincidences and 8 counts in the zero-delay peak; the ± 0.054 uncertainty is therefore statistical and would shrink as $1/\sqrt{N}$ with longer integration. The thermometry has a larger problem: three consecutive scans with byte-identical parameters give $\eta^2 n_{\text{excitation}}$ spanning a factor of four, and the pulse length is scanned monotonically rather than in random order, so slow drift aliases onto the time axis. We have inflated the error bar by the observed scatter, but the honest statement is that $n_{\text{excitation}} = 2.74 \pm 1.12$ is an upper bound with a systematic tail; a randomised-order scan interleaved with red/blue sideband ratios would settle it.

Sequence timing. The 1 μs dwell specified for this study is shorter than the 1.7 μs photon generation window actually used in the HBT measurement. A production sequence should use a ~ 2 μs dwell; since the residual excitation is set by the moves and not by the dwells, and since compensation makes the dwell length irrelevant to the filter settling, this costs 8 μs of the 500 μs window and nothing else.

VI. CONCLUSION

For a 100 km heralded ion–ion link the limiting timescale is not any laser, detector or fibre property but the 500 μs it takes light to tell the nodes what happened. That caps one emitter at 2000.000 attempts per second. Filling the dead time with other ions is the cheapest available improvement, and of the five multiplexing schemes we compared, walking the chain past a single fixed beam is the one that requires no new optics, preserves a single diffraction-limited collection mode, and composes with every future upgrade.

We have designed that scheme for our trap from first principles. The measured electrode geometry gives $\omega_r/2\pi = 3.222372$ MHz and $\omega_z/2\pi = 112.090$ kHz, a nine-ion chain 102.587 μm long, and an 88 μs sequence that addresses every ion. The technically interesting result is negative and then positive: invariant-based inverse engineering alone is *not* sufficient, because the sealed 100 kHz in-vacuum filter between the DAC and the electrodes converts a nominally perfect waveform into 990.077 quanta of centre-of-mass excitation, and no accessible move duration between 2 and 20 μs repairs it. Composing the shortcut with an exact inverse filter does repair it, completely and cheaply — 22 orders of magnitude of excitation removed for 2.7 μm of extra control excursion. We regard the general lesson as the useful one: a shortcut to adiabaticity is a statement about the potential the ion sees, and it is only as good as the transfer function between the waveform you compute and the potential you get.

Experimentally the multiplexed chain already behaves as advertised: $g^{(2)}(0) = 0.147 \pm 0.054$ confirms single-photon emission from one addressed ion at a time, and Rabi thermometry gives $n_{\text{excitation}} = 2.74 \pm 1.12$ quanta, deep below the Doppler limit. The remaining step is the one the link cannot do without: replacing the 397 nm proof-of-concept photon with an 854 nm network photon converted to 1550 nm, which is what turns a 3000 dB budget into an 18 dB one. With that in place, transport multiplexing takes the projected pair rate over 100 km from one every 18 seconds to roughly one every two seconds.

DATA AND CODE AVAILABILITY

The four control waveforms accompany this manuscript: 1.csv (single-ion 100 $\mu\text{m}/10$ μs transport, 10,000 samples), 2.csv (ideal nine-ion 88 μs addressing sequence, 10,000 samples), 3.csv (residual COM excitation versus move duration, 2–20 μs), and 4.csv (the inverse-filtered control that is actually programmed, same span and grid as 2.csv). Column 2 of 1.csv, 2.csv is the trap-centre position along the trap axis; for 4.csv it is the commanded control, whose filtered image is the trap centre. Key scalar results are collected in result.md.

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